Course: BCA-601 (Sixth Semester)

Artificial Intelligence

BLOCK I – OVERVIEW OF ARTIFICIAL INTELLIGENCE (Marks 10)

Q1. Define Artificial Intelligence (AI) and explain its importance.

Answer:
Introduction:
Artificial Intelligence (AI) is the **branch of computer science that aims to create machines capable of intelligent behavior**. AI enables machines to **perceive, reason, learn, and make decisions** similar to humans.

Importance:

- Solves complex problems faster than humans.
- Automates repetitive tasks, improving efficiency.
- Enables advanced applications like robotics, expert systems, and natural language processing.

Example:

- Chatbots like virtual assistants that understand human queries.
- Self-driving cars making real-time decisions.

Advantages:

1. Enhances productivity and efficiency.
2. Reduces human error in critical tasks.
3. Enables decision-making in complex domains.

Limitations:

1. High development cost.
2. Requires huge data and computational power.
3. Cannot fully replicate human intuition and creativity.

Q2. Explain the related field of AI: Machine Learning.

Answer:
Introduction:

Machine Learning (ML) is a **subset of AI** that enables systems to **learn from data and improve performance without explicit programming**.

Explanation:

- Uses **algorithms and statistical models** to identify patterns.
- Types:
  - **Supervised Learning:** Learn from labeled data.
  - **Unsupervised Learning:** Discover hidden patterns in unlabeled data.
  - **Reinforcement Learning:** Learn by trial and error using rewards.

Example:

- Predicting stock prices using historical data.
- Email spam filters learning from user feedback.

Advantages:

- Adapts to new situations.
- Can handle large datasets efficiently.

Limitations:

- Requires high-quality data.
- Can overfit or underfit if not designed properly.

Q3. Explain AI techniques with examples.

Answer:
Introduction:
AI techniques are **approaches used to solve problems using intelligent systems**. They can be **symbolic or sub-symbolic**.

Techniques:

1. **Search-based techniques:** Use algorithms to explore possible solutions.
2. **Knowledge-based techniques:** Expert systems using rules and facts.
3. **Probabilistic reasoning:** Bayesian networks for uncertainty.
4. **Learning-based techniques:** Machine learning, neural networks.

Example:

- Expert system for medical diagnosis using if-then rules.
- Chess-playing AI using search algorithms.

Advantages:

- Solves complex real-world problems.
- Provides automation and decision support.

Limitations:

- Requires accurate models or rules.
- Computationally intensive for large problems.

Q4. Explain criteria for success in AI systems.

Answer:
Introduction:
An AI system is successful if it **achieves its goals efficiently, accurately, and reliably**.

Criteria:

1. **Correctness:** Produces correct and reliable results.
2. **Efficiency:** Solves problems within acceptable time and resource limits.
3. **Adaptability:** Learns from experience and adapts to new situations.
4. **Human Interaction:** Can interact effectively with users.

Example:

- Self-driving cars must make correct driving decisions, adapt to road conditions, and ensure passenger safety.

Advantages:

- Ensures AI solutions are practical and usable.

Limitations:

- Success depends on quality of data and algorithms.

BLOCK II – PROBLEMS, PROBLEM SPACE, AND SEARCH

Q5. Explain how a problem is defined as a state-space search.

Answer:
Introduction:
In AI, a problem is represented as a **state-space** where:

- **State:** A snapshot of the system at a point.
- **Initial state:** Starting point of the problem.
- **Goal state:** Desired solution.
- **Operators:** Actions that move from one state to another.

Example:

- Maze solving: Each cell is a state; moving left/right/up/down are operators.
- Initial state: Start cell.
- Goal state: Exit cell.

Advantages:

- Provides structured approach for problem-solving.
- Enables use of systematic search techniques.

Limitations:

- Large state spaces can make search inefficient.

Q6. Explain the characteristics of a Production System in AI.

Answer:
Introduction:
A production system is a **rule-based AI model** that uses **if-then rules** to define problem-solving logic.

Characteristics:

1. **Knowledge Base:** Contains rules (productions).
2. **Working Memory:** Stores current state information.
3. **Inference Engine:** Applies rules to reach a goal.

Example:

- Medical diagnosis system:
    - Rule: IF symptoms = fever and cough, THEN disease = flu.

Advantages:

- Simple representation of knowledge.
- Easy to modify rules.

Limitations:

- Can be inefficient for large rule sets.
- Requires complete and accurate knowledge.

Q7. Explain issues in designing search problems.

Answer:
Introduction:
Designing search problems involves **selecting states, operators, and constraints** carefully to ensure solvable and efficient search.

Issues:

1. **State-space explosion:** Too many possible states.
2. **Selection of operators:** Must be valid and minimal.
3. **Goal specification:** Clear definition is required.
4. **Efficiency:** Minimize time and memory used in search.

Example:

- Chess: Huge number of possible moves leads to state-space explosion.

Advantages:

- Clear problem representation enables systematic search.

Limitations:

- Complex problems may need heuristic approaches to reduce computation.

BLOCK III – HEURISTIC SEARCH TECHNIQUES

Q8. Explain Generate-and-Test heuristic search technique.

Answer:
Introduction:

Generate-and-Test is a **simple trial-and-error method** where solutions are generated and tested for success.

Explanation:

1. Generate a candidate solution.
2. Test if it meets goal criteria.
3. If not, generate next candidate.

Example:

- Solving 8-queens problem: Place a queen in a row and test for conflicts.

Advantages:

- Simple and easy to implement.
- Does not require complex knowledge.

Limitations:

- Inefficient for large problem spaces.
- May generate many invalid candidates.

Q9. Explain Hill-Climbing heuristic search technique.

Answer:
Introduction:
Hill-Climbing is an iterative search that **moves towards the goal by selecting the best neighboring state**.

Explanation:

- Start from initial state.
- Evaluate neighboring states using **heuristic function**.
- Move to the neighbor with highest value.

Example:

- Optimizing a route: Select path that reduces distance step by step.

Advantages:

- Simple and efficient for small problems.

Limitations:

- Can get stuck in **local maxima, plateaus, or ridges**.

Q10. Explain Best-First Search technique.

Answer:
Introduction:
Best-First Search uses **evaluation function (heuristic)** to explore the most promising nodes first.

Explanation:

- Combines **breadth-first and greedy approach**.
- Evaluation function `f(n)` decides which node to expand next.

Example:

- Pathfinding in maps: Node with shortest estimated distance to goal expanded first.

Advantages:

- Reduces unnecessary exploration.
- Efficient for large state spaces.

Limitations:

- May require large memory to store nodes.
- Quality depends on heuristic accuracy.

Q11. Explain Problem Reduction and its use in AI.

Answer:
Introduction:
Problem Reduction decomposes a **complex problem into simpler subproblems** recursively until solved.

Example:

- Tower of Hanoi: Solve `n-1` disks recursively before moving the largest disk.

Advantages:

- Breaks down complex tasks.
- Supports **divide-and-conquer approach**.

Limitations:

- Subproblem dependencies may complicate solution.

Q12. Explain Constraint Satisfaction Problems (CSP) with examples.

Answer:
Introduction:
CSP involves **finding values for variables subject to constraints**.

Example:

- Sudoku: Fill numbers in a grid so that all rows, columns, and sub-grids satisfy constraints.
- Map coloring: Assign colors to regions so that no neighboring regions have same color.

Advantages:

- Systematic approach to constraint-based problems.
- Supports AI applications like scheduling and planning.

Limitations:

- Can be computationally intensive.

BLOCK IV – PREDICATE LOGIC

Q13. Explain representing simple facts in logic with examples.

Answer:
Introduction:
Predicate logic is used to **represent facts, relationships, and rules** in AI.

Explanation:

- Facts represented as **predicates with arguments**.
- Example: `Parent(John, Mary)` → John is parent of Mary.

Advantages:

- Clear and formal representation of knowledge.

Limitations:

- Can be complex for large knowledge bases.

Q14. Explain instances and 'isa' relationships in logic.

Answer:
Introduction:
'isa' relationship represents **class-subclass or object-class relationship** in logic.

Example:

- `isa(Dog, Animal)` → Dog is an instance of Animal.
- `isa(Labrador, Dog)` → Labrador is a subclass of Dog.

Advantages:

- Helps in organizing hierarchical knowledge.

Limitations:

- Can become large and complex in deep hierarchies.

Q15. Explain computable functions and predicates in logic.

Answer:
Introduction:
Computable functions are **functions that can be evaluated algorithmically**. Predicates return **true/false values** for logical statements.

Example:

- Function: `Add(x,y) = x + y`
- Predicate: `Even(4)` returns True

Advantages:

- Supports automated reasoning.

- Enables problem-solving using logic.

Limitations:

- Complex functions may be computationally expensive.

BLOCK V – KNOWLEDGE REPRESENTATION

Q16. Define knowledge and explain the importance of knowledge in AI.

Answer:
Introduction:
Knowledge is **information about the world that can be used to make decisions or solve problems**. In AI, it is the **basis for intelligent behavior**.

Importance:

- Enables reasoning and decision-making.
- Forms the basis for **expert systems and problem-solving**.

Example:

- Medical diagnosis system uses knowledge of symptoms and diseases.

Advantages:

- Facilitates automation of complex tasks.
- Reduces human intervention.

Limitations:

- Knowledge acquisition is challenging.

Q17. Explain knowledge representation in AI.

Answer:
Introduction:
Knowledge representation is a **formal way of storing knowledge** so that AI systems can use it for reasoning and problem solving.

Techniques:

- Logical representation (propositional/predicate logic)
- Semantic networks
- Frames
- Rules and production systems

Example:

- Rule: IF fever AND cough THEN flu.

Advantages:

- Enables automated reasoning.
- Supports intelligent decision-making.

Limitations:

- Large knowledge bases can be complex.

Q18. Explain various approaches used in knowledge representation.

Answer:
Introduction:
Different approaches structure knowledge for AI reasoning.

Approaches:

1. **Logic-based:** Formal statements, inference rules.
2. **Semantic Networks:** Graph of objects and relations.
3. **Frames:** Data structures with attributes and values.
4. **Production Rules:** If-then statements for reasoning.

Example:

- Expert system for medical diagnosis using rules.

Advantages:

- Flexibility in reasoning.
- Supports complex problem-solving.

Limitations:

- May require large memory and computational power.

Q19. Explain issues in knowledge representation.

Answer:
Introduction:
Knowledge representation faces challenges in **accuracy, completeness, and efficiency**.

Issues:

- Representing uncertain or incomplete knowledge.
- Handling large-scale knowledge bases.
- Maintaining consistency.

Example:

- Expert systems must handle conflicting rules or incomplete symptoms.

Advantages:

- Awareness of issues improves system design.

Limitations:

- Complex representation can reduce system performance.

Q20. Explain applications of AI and knowledge representation.

Answer:
Introduction:
AI and knowledge representation are applied in various domains to **automate tasks and support intelligent decision-making**.

Examples:

- **Medical Diagnosis:** Expert systems suggest treatments.
- **Robotics:** Knowledge of environment guides actions.
- **Natural Language Processing:** Understand and generate human language.
- **Search Engines:** Represent knowledge of web pages to answer queries.

Advantages:

- Automates complex reasoning tasks.
- Reduces human effort and error.

Limitations:

- Requires accurate knowledge and computational resources.